Review: Equations of lines

1 Explain why the following statements are incorrect.

a "A line with an undefined slope has no equation."

b "The slope of the line with equation $7x + y = 3$ is 7."

2 A line has the equation $y = 6\left(3x - 2\right)$.

a Rewrite $y = 6\left(3x - 2\right)$ in the form $y = mx + c$.

b State the slope and y-intercept of the equation.

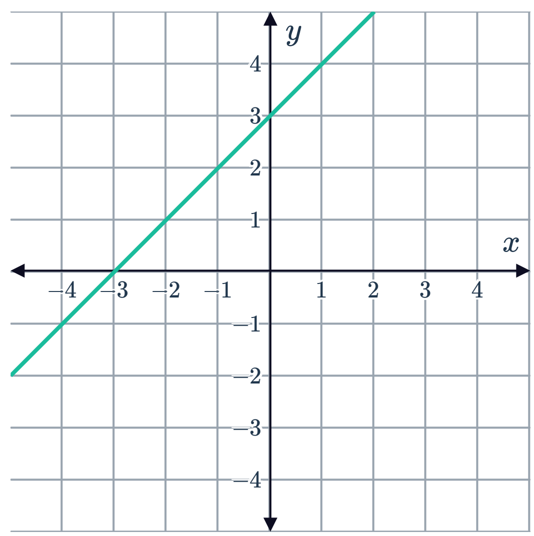
3 Write the equation of the following lines in slope-intercept form:

a A line whose slope is -8 and crosses the y-axis at -9.

b A line that has a slope equal to -7, and cuts the y-axis at 7.

c A line that passes through the point $A\left(-2, -9\right)$ and has a slope of -2.

4 The following is a graph of $y = mx + b$. Sketch the graph of y when b is decreased by 5.



5 Line L_1 has the following equation: $y = x - 2$

a Find the y-value of the y-intercept of the line.

b Find the x-value of the x-intercept of the line.

c Find the y-coordinate of a point that has an x-coordinate of -5.

d Sketch the line $y = x - 2$ on the coordinate plane.

6 Express the following equations in standard [redacted].

a $y-5x=5$

b $y=2x-5$

c $y=\frac{2x}{5}+4$

7 A line has slope $\frac{14}{3}$ and passes through the point $(-5,-10)$.

a Find the value of b for this line.

b Write the equation of the line in standard form.

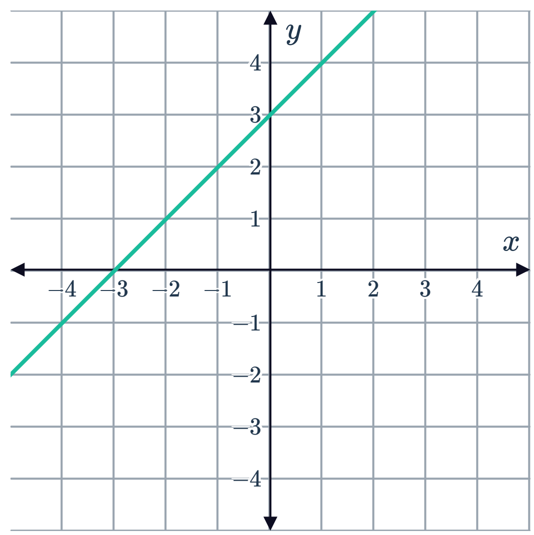
8 Determine whether the following could be the equation of the graphed line.

a $2x-2y=6$

b $2x-2y=-6$

c $2x+2y=6$

d $2x+2y=-6$



9 $A(-1,4)$ is a point on a straight line. If the slope of the line is 4, determine whether the following statements are correct.

a $\frac{x+1}{y-4}=4$

b $\frac{y-4}{x+1}=4$

c $\frac{y+1}{x-4}=4$

d $\frac{4-y}{-1-x}=4$

10 A line passes through the points $(3,-5)$ and $(-7,2)$.

a Find the slope of the line.

b Find the equation of the line in slope-intercept or standard form.

11 Use the two point formula to derive the equation of the line that passes through the points $A(8,1)$ and $B\left(-2,\frac{1}{7}\right)$. Give your answer in standard form.

12 A line has an x-intercept of 8 and a y-interc[redacted] 5.
By considering the intercept form of a straight line, write the equation of the line in the standard form.

13

a Find the equation, in standard form, of the line that passes through $A\left(7,-14\right)$ and $B\left(-7,-15\right)$.

b Find the x-coordinate of the point of intersection of the line that goes through A and B, and the line $y=x-8$.

c Hence, find the y-coordinate of the point of intersection.

Additional questions

14 For each of the following linear equations:

i State the value of m, the slope.

ii State the value of b, the y-intercept.

a $y=6x+4$

b $y=-3x-8$

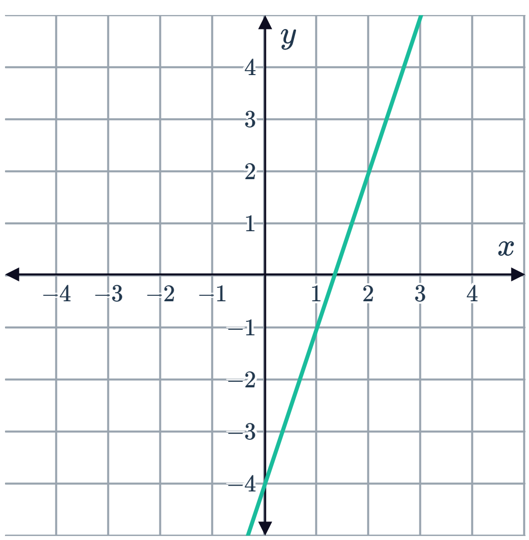
15 For each of the following graphs:

i Determine the slope of the line.

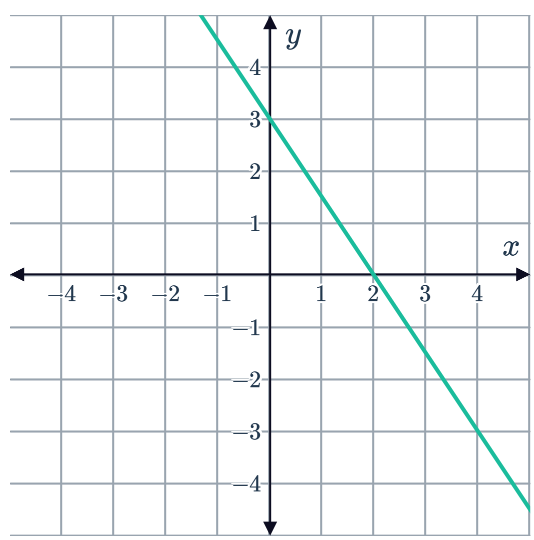
ii Determine the y-intercept of the line.

iii Determine the equation of the line.

a



b



16 Consider the following linear equations:

- Line 1: $y = 2 + 6x$
- Line 2: $y = 5x + 6$
- Line 3: $y = 6x$
- Line 4: $y = \frac{x}{6} + 2$
- Line 5: $y = 5 - 6x$
- Line 6: $y = 6x - 2$

State the three lines that have the same slope.

17 For each of the following lines, state whether the following lines have a y-intercept or not. If the line does have a y-intercept, state the coordinates of it.

a $y = 2$ **b** $5x = 4y$ **c** $y = -4x + 1$ **d** $x = 1$

18 Suppose the Line L_1 has the equation $y = \dfrac{-9x + 3}{3}$.

a Simplify L_1 to the form $y = mx + b$.

b State the slope of the line.

c Find the y-value of the y-intercept of the line.

19 Determine whether each of the following lines has the same slope as $y = 6x + 3$ or not.

a $y = 6x$ **b** $y = 3x + 3$ **c** $12x - 2y = 3$ **d** $18x + 3y = 0$

20 Find the equation of the lines matching the following descriptions:

a A line that has the same slope as $y = 2 - 4x$ and the same y-intercept as $y = -7x - 3$.

b A line whose slope is $\dfrac{2}{3}$ and goes through the point $(0, 5)$.

c A line that is horizontal and goes through the point $(9, -2)$.

d A line that is vertical and goes through the point $(11, -9)$.

21 Sketch the following lines on a coordinate plane:

a The line that has a y-intercept of -4 and whose slope is $-\dfrac{4}{3}$

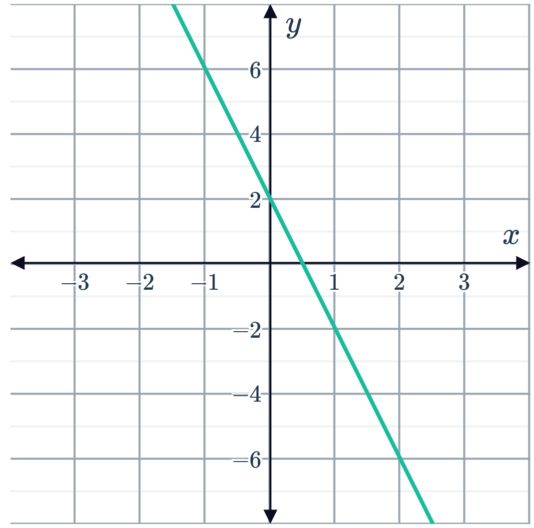
b The line $y = 2x + 4$

c The line $y = -2x + 4$

d The line $y = \dfrac{1}{2}x - 1$

e The line $x = -3$

22 Consider the following line:



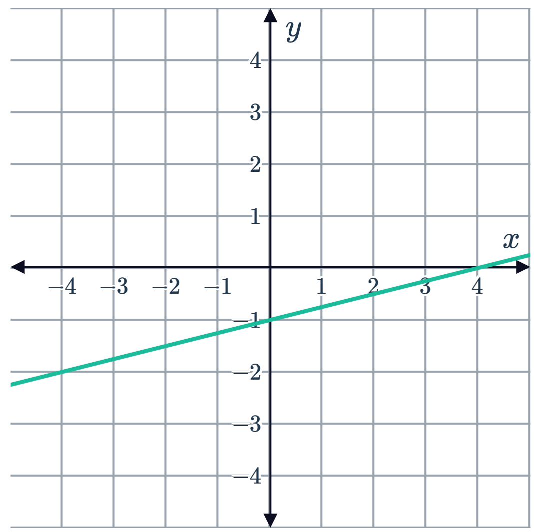
a. Copy and complete the following table of values.

x	-1	0	1	2
y				

b. State the values of the slope, m and the y-intercept b.

c. Write the linear equation expressing the relationship between x and y.

23 Consider the following line:



a. State the values of the slope, m and the y-intercept b.

b. Write the linear equation expressing the relationship between x and y.

24 For the following lines passing through the given two points:

i. Find the slope of the line.

ii. Find the equation of the line in the form $y = mx + b$.

a. $(0, 5)$ and $(2, 9)$

b. $(-3, 9)$ and $(4, 2)$

c. $(-6, 8)$ and $(6, 0)$

d. $(-7, 3)$ and $(-7, -2)$

25 State the standard form of a linear equation.

26 State which of the following linear equations are in standard form:

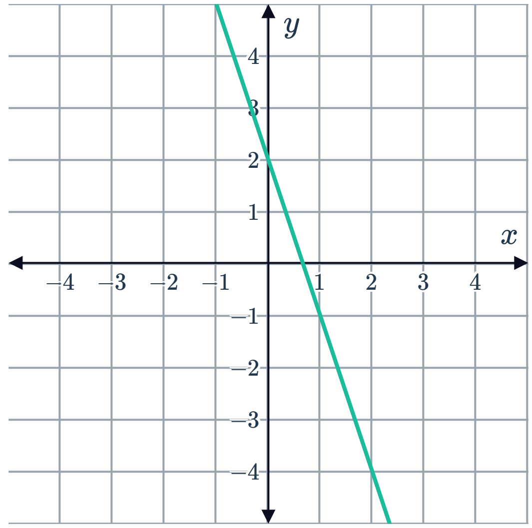
a. $2x - 5y = 0$

b. $x = 10$

c. $y = \frac{1}{[illegible]}x$ [illegible] 5

d. $3x + \frac{7}{[illegible]}y = 11$

27 Consider the following line:



a State the y-intercept.

b Find the slope, m, of the line.

c Find the equation of the line in the form $y = mx + b$.

d Rewrite the equation of the line in standard form.

28 State the point-slope form of a linear equation.

29 For each of the following equations, state the form they are written in:

a $y = -4x + 5$

b $y - 1 = 5(x - 4)$

c $5x - 3y = 8$

30 Write the equation of the following lines in point-slope form:

a A line passing through the point $(7, 1)$ has a slope of 4.

b A line passing through the point $(4, 0)$ has a slope of -5.

c A line passing through the point $(1, 6)$ has a slope of 0.

31 For each of the following lines:

i Find the slope, m, of the line.

ii Write the equation of the line in point-slope form.

a A line passes through the two points $(7, 6)$ and $(9, 12)$.

b A line passes through the two points $(4, -8)$ and $(2, 0)$.

32 A line has slope $-\frac{3}{2}$ and passes through the [redacted]t $(2, -2)$.

a Find the value of b for this line.

b Hence, write the equation of the line in slope-intercept form.

c Derive the same equation by using the point-slope formula.

d Sketch the graph of the line.